Data USB and UART bridge

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J1
USB4920
VBUS
VBUS
VBUS
VBUS
CC1
CC2
D-
D-
D+
D+
SBU1
SBU2
SHIELD
GND
SH1
A1
A4
A9
B4
B9
A5
B5
A7
B7
A6
B6
A8
B8
GND

F1
1.2A quick acting
PWR_FLAG
VBUS
D4
P4SMAJ5.0A
GND
R2
5.1k
R3
5.1k
GND
D-
D+
D1
D5V0L2B3SO-7
GND

U1
CP2102-GM
GND_2
DTR
DSR
TXD
RXD
RTS
CTS
NC_10
DCD
RI
GND_1
D+
D-
VDD
REGIN
NC_9
NC_8
NC_7
NC_/_VPP
NC_6
NC_5
NC_4
VBUS
RST
NC_1
SUSPEND
SUSPEND
NC_2
NC_3
GND
TX
RX
R5
470
R6
470
R4
4.7k
VDD
PWR_FLAG
C1
1u
C2
100n
C3
100n
C4
1u
D3
STPS1L30A
VBUS

Reset and boot pin configuration

U3
IO0
RTS
R10
12k
R13
12k
DTR
EN
BC847BDW1T1G
+3V3
10k
R14
IO0
IO2
10k
R15
GND
+3V3
R9
10k
EN
C7
1u
SW2
D5
ESD5Z3V3
GND
IO0
SW3
D6
ESD5Z3V3
GND

ESP32 module

+3V3
C10
100n
C11
1u
C12
10u
GND
U5
ESP32-WROVER
EN
EN/CHIP_PU
3V3
32K_XP/GPIO32/ADC1_CH4
32K_XN/GPIO33/ADC1_CH5
RV1
MTDI/GPIO12/ADC2_CH5
MTCK/GPIO13/ADC2_CH4
BZ
MTMS/GPIO14/ADC2_CH6
SW2
MTDO/GPIO15/ADC2_CH3
GPIO34/ADC1_CH6
VCC_DVDD
GPIO35/ADC1_CH7
SENSOR_VP/GPIO36/ADC1_CH0
SENSOR_VN/GPIO39/ADC1_CH3
U0TXD/GPIO1
U0RXD/GPIO3
GPIO0/BOOT/ADC2_CH1
ADC2_CH2/GPIO2
ADC2_CH0/GPIO4
GPIO5
GPIO18
GPIO19
GPIO21
GPIO22
GPIO23
DAC_1/ADC2_CH8/GPIO25
DAC_2/ADC2_CH9/GPIO26
ADC2_CH7/GPIO27
GND_THERMAL
GND
RX
TX
IO0
IO2
SW3
SW4
SW1
LEDR
SW5
ESP32-WROVER-E

Input commands

RV1
10k
10k
+3V3
PWR_FLAG
SW1
SW4
SW2
SW5
SW3
SW6
SW4
SW7
SW5
SW8
GND

Low battery indicator

+3V3
R18
62
D7
LED_R_SMD
LEDR
R16
22
R17
10k
Q2
2N7002
GND

Buzzer

+3V3
R21
3.9k
BZ1
Buzzer
BZ
R19
22
R20
10k
Q3
2N7002
GND

Charging indicator

VBAT
D8
LED_B_SMD
R22
1k
CHG

Battery and holder

VBAT
BT1
1S1P Li-Ion battery
GND

Battery output enable

VBAT GND
SW1
200MSP1T1B2M6QE
MH2
MH1
GND
VBUS
Q1
MMFTP3401
G
VCC
D2
PMEG3010BER
PWR_FLAG
R1
100k
GND

Battery charge circuit

U2
BQ21040DBVR
VBUS
VIN
ISET
OUT
CHG
GND
TS
VBAT
CHG
C5
1u
R7
1.2k
R8
10k
C6
1u
GND

DC-DC converter

U4
LD39100PU33R
VCC
VIN
EN
NC
VOUT
PG
GND
+3V3
C8
1u
C9
22u
GND

Supply voltage level detection

VCC
R11
1k
VCC_DVDD
R12
1k
GND

Jakov Begović
Sheet: /
File: Kontroler.kicad_sch
Title: Robot upravljan kontrolerom v2 – kontroler
Size: User
Date: 2026-10-26
Rev: v1
KiCad E.D.A. 10.0.6
Id: 1/1